

Roll No.....

END SEMESTER EXAMINATION JUNE-2025

Name of the Program: B. Tech Petroleum Engineering
Name of the Course: Petroleum Drilling Engineering II

Semester: 6th
Course Code: TPE 604

Time: 3 Hrs

Maximum Marks: 100

Notes:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a,b, & c in each main questions
- (iii) Total marks in each main question are twenty.
- (iv) Each questions carries 10 marks.

Q1	(20 marks)	
(a)	What is directional drilling, and how does it differ from vertical drilling? What are the applications of directional drilling?	CO1
(b)	What are the parameters to be consider in directional well planning?	CO1
(c)	What is dogleg severity (DLS) in directional drilling? What causes doglegs to form during directional drilling?	CO1
Q2	(20 marks)	
(a)	What are well profile design considerations in horizontal well?	CO3
(b)	How do torque and drag behave differently in horizontal drilling compared to vertical drilling?	CO3
(c)	Define the kick-off point (KOP) and explain its role in trajectory design.	CO2
Q3	(20 marks)	
(a)	How does differential sticking occur, and what are the common prevention techniques?	CO4
(b)	What is keyseating, and how can it be prevented during directional drilling?	CO4
(c)	What is a kick, and what steps are taken to control it during drilling?	CO4
Q4	(20 marks)	
(a)	What are the main types of casing strings used in oil and gas wells? Why is formation pressure a critical factor in casing design?	CO5
(b)	What are the primary design criteria considered in casing design? What is burst pressure, and what factors influence burst failure in casing?	CO5
(c)	How does well trajectory (vertical, directional, or horizontal) affect casing program planning?	CO5
Q5	(20 marks)	
(a)	What is a drill string, and what are its main components?	CO6

(b)	How does hole geometry (depth, diameter, inclination) influence drill string design?	CO6
(c)	What factors influence the selection of drill pipe in a drilling operation? What is the difference between standard and heavy weight drill pipe (HWDP)?	CO6